

Shahjalal University of Science & Technology, Sylhet
 Department of Computer Science & Engineering
Final Examination, 2nd Year 1st Semester, 2018
 Course No: CSE 233 Course Title: **Object Oriented Programming Language**
 Credit:3, Full Marks:100 Time: 3 Hours

(Answer any four questions taking two from each group)

GROUP-A

1. a. Define True/False. If false, give correct answer. 5
- The range of character data type in Java is -32768 to +32767.
 - inti = 212L; is a valid statement in Java.
 - In Java, *Double* is a primitive data type.
 - In Java, the size of a one-dimensional array can be determined by *length()* method.
 - int[] arr = {10, 20, 30}; is a valid statement in java.
- b. For the following literals, determine valid or invalid. If valid, write the data type: 10
- | | |
|--------------|---------------|
| i. 99.5 | vi. 9_9.5_3f |
| ii. 20L | vii. "22.38" |
| iii. 0b10_11 | viii. 0x5.5e5 |
| iv. 0b1011_ | ix. 0x5.5p5 |
| v. 1_2_3 | x. _1_23 |
- c. Check the following java program carefully. Do you think there is any compilation error? If there is find the errors, correct it using line number and then write the output. For correcting you cannot insert new statement or delete a statement. If there is no error then write only output. 10
- ```

01 public class OpExam {
02 public static void main (String[] args) {
03 int a = 317, b = 65, c = 177;
04 a = --b + c++;
05 System.out.println("1)" + ++a + " " + b + " " + c);
06 a = 21; b = 19;
07 System.out.println("2)" + (a^b));
08 System.out.println("3)" + (~b));
09 byte i = 5; a = i << 3;
10 System.out.println("4)" + a);
11 i = -1; a = i >> 5;
12 System.out.println("5)" + a);
13 a = i >>> 28; System.out.println("6)" + a);
14 a = 87; a *= 8; System.out.println("7)" + a);
15 boolean d = true, e = false;
16 d |= e; System.out.println("8)" + d);
17 d &= e; System.out.println("9)" + d);
18 System.out.println("10)" + ((a%4==0 && e) ? "Close" : "End"));
19 }
20 }

```
2. a. Provide two differences between procedural and object oriented programming? 3+2
- What if main method is written as "static public void" instead of "public static void"?
- b. What is the best approach if frequent string concatenation is performed? Write the difference between StringBuffer and StringBuilder class? 3+2
- c. Write the output of the following code in java: 10
- ```

public class StringDemo {
    public static void main(String args[]){
        String s1 = "Java World";
        String s2 = "Java ";
        String s3 = "Programming";
        System.out.println(" " + s1.substring(5));
        System.out.println(" " + s1);
        s1 = s1.substring(5);
        System.out.println(" " + s1);
        System.out.println(" " + s2.concat(s1).toLowerCase());
        System.out.println(" " + s2+"$");
        System.out.println(" " + s2.length());
        System.out.println(" "+"java".equalsIgnoreCase(s2.trim()));
        System.out.println(" " + "java".equals(s2.trim()));
        System.out.println(" " + s3.charAt(3));
        s2 = s2.replace("a", "A");
        System.out.println(" " + s2.toUpperCase().equals("JAVA"));
    }
}
    
```
- d. Despite being a popular language, what is the limitation of C language? Write on *final* variable, *final* method and *final* class with example. 2+3

3. a. When does garbage collector in Java works? Write three restrictions of static methods. 2+3
- b. Write down the usage and restriction of *this()* in a class. What happens if Java allows multiple inheritance? Show an example. 2+3
- c. What do you know about *Dynamic Method Dispatch* in Java? 3
- d. Check the following code carefully. Write the output of the following codes with explanation to your answer. If there is any error, specify it with reason 12

```

class Animal{
    void jump(){
        System.out.println("Animal");
    }
    void jump(double a){
        System.out.println("Animal:"+a);
    }
}
class Cat extends Animal{
    void jump(int a){
        System.out.println("Cat");
    }
}
class Horse extends Animal{
    void jump(int a){
        System.out.println("Horse");
        super.jump(a);
    }
}
public class Circus {
    public static void main(String args[]){
        Animal cat = new Cat();
        Horse horse = new Horse();
        cat.jump();
        horse.jump(10.1);
        horse.jump(10);
    }
}

```

GROUP-B

4. a. Define True/False. If false give correct answer. 5
- The default layout manager of Java Swing frame is flowlayout.
 - Default (access modifier) members of a class can only be accessible subclass of different package.
 - Unlike *Hashtable*, *HashMap* is synchronized.
 - this()* can be called from any non-static method in a class.
 - Super class object can be referenced by subclass variable.

- b. Define the class *Point*, *Shape*, *Rectangle* and *Circle* for the following program. All member variables of these classes should be private. 4*2.5

```

01 public class ObjectToClass {
02     public static void main (String[] args) {
03         Point p = new Point(1,2);
04         System.out.println("Point:"+p.getX()+" "+p.getY());
05         Rectangle r = new Rectangle(p,20.0,40.0);
06         System.out.println("R:"+r.getHeight()+" "+r.getWidth());
07         System.out.println("R-Area:"+r.area());
08         r.setWidth(8.0); r.setHeight(6.0);
09         Shape s = r; System.out.println("S-Area:"+s.area());
10         System.out.println("S-Point:"+s.getPoint().getX()
11                                 +", "+s.getPoint().getY());
12         p.setX(4); p.setY(6);
13         Circle c = new Circle(p,5.0);
14         System.out.println("C:"+c.getRadius()+" "+C-Area:"+c.area
15         c.setRadius(4.0);
16         s = c; System.out.println("S-Area:"+s.area());
17     }
18 }

```

Output of this program:

Point:1,2
 R:40.0,20.0
 R-Area:800.0
 S-Area:48.0
 S-Point:1,2
 C:5.0,C-Area:78.5
 S-Area:50.24

- c. Write the output of the following program with explanation?

10

```
public class TestException {
    void test() {
        try {
            String s = null;
            System.out.println("1");
            try {
                System.out.println(s.length());
            } catch (NullPointerException e) {
                System.out.println("inner");
                throw e;
            }
            System.out.println("2");
        } catch (NullPointerException e) {
            System.out.println("outer");
            return;
        } finally {
            System.out.println("3");
        }
        System.out.println("4");
    }
    public static void main(String args[]) {
        TestException ob = new TestException();
        ob.test();
    }
}
```

1. inner
 2. outer
 3. 4

5. a. Define True/False. If false give correct answer.

5

- Java does not support multiple inheritance for Interfaces.
- All methods in an Interface must be abstract and public.
- RuntimeException* is a checked exception.
- An abstract class don't have any constructor.
- Interfaces may have abstract constructor.

- b. Differentiate between Checked Exception and Unchecked Exception? What happens if a class partially implements an interface?

3+2

- c. Write down the output of the following program:

10

```
abstract class A {
    A(int a) { System.out.println("A(int)"); }
    A() { System.out.println("A()"); }
    abstract void x();
    void y() {
        System.out.println("A.y()");
    }
}

class B extends A {
    void x() {
        System.out.println("B.x()");
    }
    void y() {
        System.out.println("B.y()");
    }
    void z() {
        System.out.println("B.z()");
        super.y(); y();
    }
}

public class TestDemo {
    public static void main(String[] args) {
        B b = new B(); b.z();
    }
}
```

- d. Write on throws keyword in Java? How can you create a new custom unchecked Exception class.

3+2

6. a. Differentiate between "protected" and "default" access specifiers. 3
- b. Write a Java GUI program in which you are required to extend JFrame. Create three text fields to take input of height, width, depth for a Box object. Clicking on 'Create' button will create a new Box object and add it into an ArrayList. Clicking on 'View' button will print the total ArrayList into console output using for-each version of for loop. Add necessary labels for the input textfields. 12
- c. How can you make your java method thread safe? What is the purpose of join() method in Thread class? 2+3
- d. What is the difference between Listener and Adapters in Java? Which one is better? 3+2

[Answer any ten questions taking five from each group]

Group A

- Q1 a) Graphically define the notations $O(n)$, $\Omega(n)$ and $\Theta(n)$. 2
b) For each of the following recurrences, give an expression for the runtime $T(n)$ if the recurrence can be solved with the Master Theorem. Otherwise, indicate that the Master Theorem does not apply. 2*4
i) $T(n) = 3T(n/2) + n^2$
ii) $T(n) = T(n/2) + 2^n$
iii) $T(n) = 16T(n/2) + n$
iv) $T(n) = 2T(n/2) + n/\log(n)$

- Q2 a) Let the data 63, 71, 82, 64, 56, 40, 55, 74 are stored in $A[1]$ to $A[8]$. Now show each step to sort those data using Merge Sort algorithm. 8
b) What will be the best, average and worst case for MergeSort technique while sorting N numbers? 2

- Q3 Demonstrate how divide and conquer technique is used in this algorithm to find the key element $k = -7$ for the following entries: 10
-15, -5, 0, 9, 14, 23, 46, 54, 83, 113, 123, 141, 144, 151

- Q4 Apply the greedy activity selection algorithm to the following set of activities,

Activity	A	B	C	D	E	F	G	H	I
Start Time	1	1	1	2	3	4	5	6	7
Finish Time	2	3	4	5	7	9	6	8	9

- a) Show all steps to find the selected activities. 7
b) What is the running time of this algorithm? 1
c) Have you find the optimal solution? 2
i. If your answer is yes, state why you found the optimal solution.
ii. If your answer is no, then propose another way to find the optimal solution.

- Q5 Find an optimal solution to the following knapsack instance using dynamic programming: 10

Number of Objects $N=6$, Maximum capacity $M=15$ and

P_i	10	5	15	7	8	18
W_i	3	4	3	5	2	6

Where P_i and W_i denotes the price and weight of the i -th product respectively.

- Q6 Five matrices are given with their dimension:

$$A1 = 5 * 10 \quad A2 = 10 * 15 \quad A3 = 15 * 5$$

$$A4 = 5 * 100 \quad A5 = 100 * 20$$

Using the Dynamic programming approach for Matrix Chain Multiplication, Find

- i) The Optimal Order of Multiplication. 2
ii) The minimum number of multiplication required for the multiplication. 8
- Q7 Find the minimum edit distance between 'INTENTION' and 'EXECUTION'. Show each step to describe your algorithm. 10

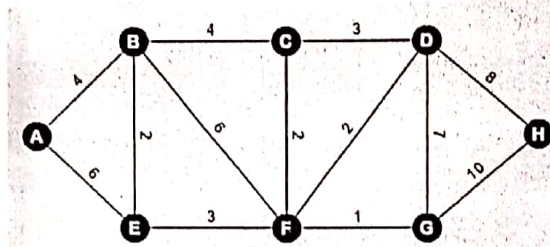
Group B

- Q1 Answer the following questions in short (Any Five): 2*5
a) What is the Best Case, Worst Case and Average Case Complexity of Quick Sort Algorithm?

- b) Give an Example of Substitution Method.
- c) What is Asymptotic Complexity?
- d) Write at least Three Purpose of Hashing.
- e) What is Geometric Hashing?
- f) What do you mean by Optimal Solution?
- g) What is Recursion Tree Method?

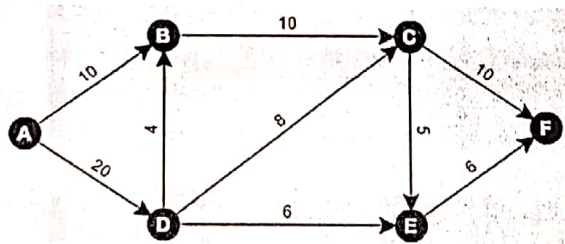
- Q2 a) What are the differences between backtracking and branch-and-bound? 2
 b) Find the solution of 5-queen problem using backtracking method. 8

- Q3 Find the shortest path from vertex A to H using Dijkstra's Algorithm. Show all steps. 10

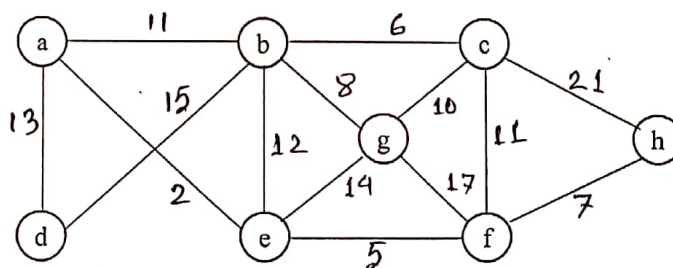


- Q4 Construct a Graph with 8 Nodes and 28 Directed Edges. Now, find first 7 Fibonacci Primes and use each of them four times to give the weight of the edges. Then take a graph algorithm which was not in the syllabus of any past courses but in this course. Then exemplify that algorithm with this graph. 10

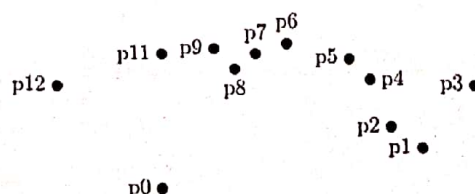
- Q5 Consider the following network. Run the Ford-Fulkerson algorithm to compute a max-flow. For each iteration give the residual network and mark the path you choose for augmentation. 10



- Q6 Consider the weighted graph in the following figure. Apply Kruskal's algorithm to find out the minimum-cost spanning tree. Show the steps. 10

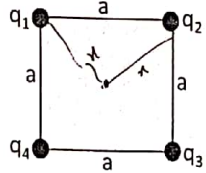


- Q7 a) Compare the Dijkstra's algorithm and Bellman-Ford algorithm in finding shortest path for a graph. 4
 b) Show all the steps to run Graham's scan on the following points to form a convex-hull. 6





[Answer any five questions from the following. The figures on the right margin indicate full marks.]

1. (a) State Huygen's principle of propagation of waves. 2
 (b) Define interference of light. Find the fringe-width in Young's double slit experiment. 2+6
 (c) The separation between the slits is 2.5 mm and the distance of the screen from the slits is 50 cm in a Young's double slit experiment. If a sodium light of $\lambda = 5890 \text{ \AA}$ is used in the experiment, calculate (i) the angular position of the first maxima and (ii) the fringe-width. 4
2. (a) What are Newton's rings and how are they formed? 4
 (b) Explain Michelson interferometer and describe the formation of fringes in it. 7
 (c) In a Michelson interferometer 200 fringes cross the field of view when the movable mirror is displaced through 0.0589 mm. Calculate the wavelength of monochromatic light used. 3
3. (a) What do you mean by diffraction of light? Obtain the expression of intensity pattern due to a single-slit Fraunhofer diffraction. Hence draw the intensity distribution curve. 2+7+1
 (b) What is diffraction grating? Calculate the possible order of spectra with a plane transmission grating having 18,000 lines per inch when a light of wavelength $4500 \text{ \AA}$ is used. 1+3
4. (a) What do you mean by polarization of light? Mention some methods of polarization. 2+2
 (b) State and prove Brewster's law of polarization. 2+3
 (c) Define specific rotation of an optically active material. 1+4
 A 200 mm long tube containing 48 cm^3 of sugar solution produces an optical rotation of 11° . If the specific rotation of the sugar solution is 66° , calculate the concentration and quantity of the sugar contained in the tube in the form of a solution.
5. (a) State Gauss's law. Deduce Coulomb's law from Gauss's law. 2+3
 (b) If a hypothetical cylinder of radius R immersed in a uniform electric field $\vec{E}$, the cylinder axis being parallel to the field. Calculate the flux ϕ_E for this closed surface. 5
 (c) What is the potential at the center of the square of the figure? 4
 Where the charges are, $q_1 = 1.0 \times 10^{-8} \text{ C}$, $q_2 = -2.0 \times 10^{-8} \text{ C}$, $q_3 = 3.0 \times 10^{-8} \text{ C}$, $q_4 = 2.0 \times 10^{-8} \text{ C}$ and $a = 1.0 \text{ meter}$.
- 
6. (a) When does magnetic force arise between two charges and when does a charge feel magnetic force? 1+1
 (b) Find the magnetic force on a current carrying conductor placed in a static magnetic field. 4
 (c) What is Hall effect? How can the sign and density of the carrier be determined from the Hall effect? 2+3
 (d) A strip of copper $150 \mu\text{m}$ thick is placed in a magnetic field of magnitude 0.65 T . When a current of $23 \mu\text{A}$ is flowed through the strip, the Hall potential difference across the width of the strip is found to be $7.5 \mu\text{V}$. What is the density of charge carrier of the copper strip? 3
7. (a) Write down Maxwell's equation in integral and differential form. 4
 (b) What is continuity equation? How did Maxwell fix Ampere's law? 4
 (c) How can one prove that light is an electromagnetic wave? 6
8. (a) Mention the various series of the hydrogen spectrum. 2
 (b) What is photoelectric effect? Write Einstein's photoelectric equation. 2+1
 (c) What is radioactive decay? Derive radioactive decay law. 2+3
 (d) What is matter wave? Find the de Broglie wavelength of an electron accelerating through a potential difference of V volts. 1+3

Shahjalal University of Science & Technology, Sylhet
Department of Statistics
BSc (Honours) 2nd Year 1st Semester Examination, 2018
Course No.: STA2021 for CSE (Basic Statistics and Probability)
Full Marks: 70 Time: 3 Hours Credit: 4

[Answer any FIVE questions taking at least two from each group. Figures in the right margin are assigned marks distribution. All parts of a question must be answered sequentially]

Group: A

1. (a) What do you mean by frequency distribution? Define different scale of measurement of data set with example. (3)

(b) The data on number of allocated computer for all 56 department in a university for the year 2018 are:

10 22 16 12 8 3 15 8 14 9 11 9 16 10 14 15 18 25 5 6 20 8 10 15 14 10 10 10
9 9 14 12 11 11 2 10 11 5 7 9 10 14 20 19 11 7 12 12 11 9 8 7 10 10 6 15

- i. Prepare a frequency distribution table along with class boundaries. (5)

- ii. Calculate percentage for the respective group. (3)

- iii. Draw more than type ogive curve. (3)

2. (a) Define measures of central tendency. Write down the calculating formulas of arithmetic mean, median and mode for grouped data. (4)

- (b) State the properties of AM. Show that sum of squares of deviation from mean is minimum. For $n = 4$, prove that $AM \geq GM \geq HM$. (4)

- (c) Calculate AM, median and mode from the following frequency distribution: (6)

Class Interval	:	0-2	0-4	0-6	0-8	0-10
Frequency	:	1	5	11	14	16

3. (a) Define i) moments, ii) skewness, and ii) kurtosis. (3)

- (b) Establish the relationship between raw and central moments. (4)

- (c) Given the following frequency distribution:

x_i	:	2	3	4	5	6
f_i	:	1	3	7	3	1

- i. Compute the first three central moments from the above frequency distribution. (3)

- ii. Calculate skewness for the data set and comment on it. (2)

- iii. Calculate kurtosis for the data set and comment on it. (2)

4. (a) Distinguish between correlation coefficient and regression coefficient. Find the calculating formula of correlation coefficient. What does correlation measure? (6)

- (b) State the properties of correlation coefficient. Prove that $-1 \leq r_{xy} \leq +1$. (5)

- (c) Show that correlation coefficient is the geometric mean of regression coefficients. (3)

Group: B

5. (a) Define i) classical probability, ii) random variable, iii) mutually exclusive events, and iv) equally likely events. (4)

- (b) State additive law of probability for three events. Let A , B and C are three not-mutually exclusive events. then prove that: $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$. (5)

- (c) A dice and a coin (let X and Y) are thrown simultaneously. i) Write down the sample space, ii) What is the probability of a) X is even and Y is head, and b) X is odd and Y is tail; also find the interaction of 'a' and 'b'. (5)

6. (a) Define linear regression? Draw a figure of positive and negative linear relationship with example. (3)

- (b) To study the tensile strength of a certain type of wire, the following pairs of observations were recorded, where X is the diameter in cm and Y is the mass supported in kg/cm.

X	:	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4
Y	:	14	26	50	56	42	98	82	88	134	124

- i. Plot a scatter diagram of these data and draw a free hand line to fit the data. (2)

- ii. Calculate the least-squares regression line $\hat{Y} = b_0 + b_1X$. Draw it on scatter diagram and compare it to the free hand line. (4)

- iii. Estimate the residuals. (3)

- iv. Find the predicted tensile strength for a wire with diameter 1.5 cm and interpret your result. (2)

7. (a) Define i) marginal probability, and ii) conditional probability. (3)

- (b) State the central limit theorem. (3)

- (c) The probability that a computer is affected in its main function by a virus is 0.20. Suppose this virus is set up into three computers.

- i. Draw a tree diagram for joint probabilities. (3)

- ii. Find the probability that all three are affected to this virus. (3)

- iii. Find the probability that at least one of those is not affected by virus. (2)

8. (a) What is stochastic process? Give some real life examples. (3)

- (b) Write down assumptions of the basic queuing process. Draw a birth-and-death process rate diagram. (3)

- (c) Define Markov chains with examples. Write down the properties of a Markov chain with examples. (4)

- (d) Give two examples of Markov processes representing the two classes with respect to discrete or continuous time and denumerable or continuous state space. (4)

[Answer any 5 (five) from the following Questions]

[Every part of a question must be answered sequentially, like 1(a), 1 (b), 1 (c)]

1. a. What is meant by Cost Accounting? How does cost accounting differ in technique and procedure from financial accounting? 05
(2 + 3)
- b. How cost accounting is related to management accounting? 04
- c. “Cost accounting has become an essential tool of management” – Give your opinion on the statement. 05

2. a. Define product cost and period cost with example. 04
- b. Various cost and sales data for Well food ltd. for the year 2017 are as follows: 10

Items	Taka	Items	Taka
Finished goods inventory, January 01	10,000/=	Purchase of raw materials	1,35,000/=
Finished goods inventory, Dec 31	30,000/=	Raw materials inventory, Jan 01	11,000/=
Depreciation, factory	25,000/=	Raw materials inventory, Dec 31	6,000/=
Administrative expenses	1,10,000/=	Direct labor	70,000/=
Utilities	8,000/=	Indirect labor	15,000/=
Maintenance	40,000/=	WIP inventory, January 01	17,000/=
Supplies	11,000/=	WIP inventory, December 31	30,000/=
Insurance, factory	4,000/=	sales	5,00,000/=
Depreciation, office	10,000/=	Selling expenses	100,000/=

Required:

(i) Prepare a cost of goods manufactured statement

(ii) Prepare an income statement for the year 2017.

3. a. What is meant by activity base when dealing with variable costs? Give several examples of its. 03
- b. Manager often assume a strictly linear relationship between cost and volume. How can this practice be defended in light of the fact that many costs are curvilinear? 03
- c. The administration of Azalea Hill's Hospital would like a cost formula linking the cost involved in admitting patients to the number of patients admitted during a month. The admitting department's cost and the number of patients admitted during the immediately preceding 8 (eight) months are given below – 08

Month	Number of Patients Admitted	Admitting Department Cost
May	1800	\$ 14,700/=
June	1900	\$ 15,200/=
July	1700	\$ 13,700/=
August	1600	\$ 14,000/=
September	1500	\$ 14,300/=
October	1300	\$ 13,100/=
November	1100	\$ 12,800/=
December	1500	\$ 14,600/=

Required:

(i) Use the high – low method to establish the fixed and variable components of admitting cost.

(ii) Express the fixed and variable components of admitting costs as a cost formula in the linear equation form, $Y = a + bx$.

4. a. What is the basic differences between Absorption Costing and Variable Costing? 03
- b. Are selling and administrative expenses treated as product cost or as period cost under variable costing? 03

- c. Dexture Company produces and sells a single product, a wooden hand loom for weaving small items such as scarves. Selected cost and operating data relating to the product for 2 (two) year are given below –

Selling price per unit		\$ 50/=
Manufacturing Costs:		
Variable per unit produced:		
Direct materials		\$ 11/=
Direct Labor		\$ 6/=
Variable overhead		\$ 3/=
Fixed per year		\$ 1,20,000/=
Selling and administrative Costs:		
Variable per unit sold		\$ 5/=
Fixed per year		\$ 70,000/=

	Year – 1	Year – 2
Units in beginning inventory	0	2000
Units produced during the year	10000	6000
Units sold during the year	8000	8000
Units in ending inventory	2000	0

Required:

- (i) Compute the unit product cost in each year using absorption costing.
(ii) Compute the unit product cost in each year using variable costing.

5. a. What is EOQ? What are the assumptions of EOQ?
b. What do you know about the costing procedures for scrap, spoiled goods, and defective work?
c. Following is the store ledger of Saxon co. Ltd.:

04
03
07

Date	Description	Units	Unit cost
1-Jul	Beginning inventory	500	\$ 20/=
10-Jul	Inventory purchased	800	\$ 24/=
15-Jul	Inventory Issued	900	
18-Jul	Inventory purchased	1000	\$ 25/=
25-Jul	Inventory purchased	200	\$ 30/=
29-Jul	Inventory issued	1300	

Required:

Prepare store ledger account and calculate ending inventory by using
(i) First in, first out (FIFO) method and (ii) Last in, first out (LIFO) method

6. a. What is quantity standard? What is price standard?
b. What do you understand by material price variance and material usage variance?
c. Why are separate price and quantity variances computed?
d. Xavier Company produces a single product. Variable Manufacturing overhead is applied to product on the basis of direct labor – hours. The standard cost for 1 (one) unit of product are as follows –

02
02
02
08

Direct materials: 6 ounces at \$ 0.50/= per ounce	\$ 3/=
Direct labor: 1.8 hours at \$ 10/= per hour	\$ 18/=
Variable manufacturing overhead: 1.8 hours at \$ 5/= per hour	\$ 9/=
Total standard variable cost per unit	\$ 30/=

During June, 2000 units were produced. The cost associated with June's operations were as follows –

Material purchased: 18000 ounces at \$ 0.60/= per ounce	\$ 10,800/=
Material used in production: 14000 ounces	
Direct labor: 4000 hours at \$ 9.75/= per hour	\$ 39,000/=
Total standard variable cost per unit	\$ 20,800/=

Required:

Compute the materials, labor, and variable manufacturing overhead variances.

P. T. O.

7. a. What is Static Budget? What is the difference between static budget and flexible budget? **04**
 b. Nyma repair service provides specialty repair services to the client. The company's cost budget for November 2017 appears below: **10**

Particulars	Fixed cost per month in \$	Cost per car repair in \$
Cleaning Supplies		0.80/=
Electricity	1,200/=	0.15/=
Maintenance		0.20/=
Wages and Salaries	5,000/=	0.30/=
Depreciation	6,000/=	
Rent	8,000/=	
Administrative Expenses	4,000/=	0.10/=

Company's expects to wash 9000 cars in august and to collect an average of \$ 4.90/= per car washed. But actual results are as follows:

Actual Car wash	\$ 8,900/=
Revenue	\$ 43,080/=
Expenses:	
Cleaning Supplies	\$ 7,560/=
Electricity	2,670/=
Maintenance	2,260/=
Wages and Salaries	8,500/=
Depreciation	6,000/=
Rent	8,000/=
Administrative Expenses	4,950/=
Total expenses	\$ 39,940/=
Net operating Income	\$ 3,140/=

Requirement:

- (i) Prepare a company's flexible budget
 (ii) Prepare a report showing activity variance for November 2017.
8. a. Write short notes on the followings (any four) – **(4 x 3.5) 14**
 (i) Debt Finance, (ii) Trading on Equity, (iii) Mixed Cost, (iv) Factory Overhead and
 (v) Angle of Incident.